Amazon CloudFront — Important Interview Questions & Answers

1.What is Amazon CloudFront?
Amazon CloudFront is a content delivery network service in AWS used to deliver websites, applications, videos, images, and other content to users with low latency. It uses a global network of edge locations to deliver content closer to users.
Simple ga gurthupettuko:
CloudFront ante users ki content fast ga deliver cheyyadaniki use chese AWS CDN service.
Eng: CloudFront is an AWS CDN service used to deliver content with low latency.

2.What is a CloudFront Distribution?
A CloudFront Distribution is a configuration that defines how CloudFront delivers content to users. It contains settings such as origins, behaviors, caching, and security configuration.
Simple ga gurthupettuko:
Distribution ante CloudFront content ni users ki ela deliver cheyyalo define chese configuration.
Eng: A CloudFront Distribution defines how content is delivered to users.

3.What is an Origin in CloudFront?
An Origin is the source from which CloudFront gets the content. The origin can be an S3 bucket, Application Load Balancer, EC2-based application, or another supported HTTP server.
Simple ga gurthupettuko:
Origin ante CloudFront content teesukune original source.
Eng: An Origin is the source from which CloudFront gets content.

4.What is an Origin Access Control (OAC)?
Origin Access Control allows CloudFront to securely access an S3 origin without making the bucket publicly accessible. It helps control access between CloudFront and the S3 bucket.
Simple ga gurthupettuko:
OAC ante CloudFront nundi S3 bucket ki secure access provide chese feature.
Eng: Origin Access Control provides secure access from CloudFront to an S3 origin.

5.What are CloudFront Behaviors?
Behaviors define how CloudFront handles requests for different URL path patterns. They can control settings such as caching, allowed HTTP methods, origin selection, and viewer protocol policy.
Simple ga gurthupettuko:
Behaviors ante different URL paths ki CloudFront ela behave avvali ani define chese settings.
Eng: CloudFront Behaviors define how requests are handled for different URL paths.

6.What is a Cache Policy?
A Cache Policy defines how CloudFront caches objects and which parts of a request are considered when creating cache keys. It can control settings such as headers, cookies, and query strings.
Simple ga gurthupettuko:
Cache Policy ante content ni ela cache cheyyalo and cache key ela create cheyyalo define chestundi.
Eng: A Cache Policy defines how CloudFront caches content and creates cache keys.

7.What is an Origin Request Policy?
An Origin Request Policy defines which headers, cookies, and query strings CloudFront sends to the origin when forwarding a request. These values can be sent to the origin without necessarily being included in the cache key.
Simple ga gurthupettuko:
Origin Request Policy ante request lo ye headers, cookies, query strings origin ki pampalo decide chestundi.
Eng: An Origin Request Policy controls which request values are sent to the origin.

8.What is a Response Headers Policy?
A Response Headers Policy allows CloudFront to add or modify HTTP response headers before returning the response to the viewer. It can be used for security headers, CORS headers, and other response headers.
Simple ga gurthupettuko:
Response Headers Policy ante user ki response pampetappudu required HTTP headers add cheyyadaniki use chestham.
Eng: A Response Headers Policy manages HTTP response headers sent to viewers.

9.What is SSL/TLS in CloudFront?
SSL/TLS in CloudFront is used to secure communication between viewers and CloudFront using HTTPS. CloudFront can use an SSL/TLS certificate to serve content securely over a custom domain.

Simple ga gurthupettuko:
SSL/TLS ante CloudFront and users madhya secure HTTPS communication kosam use chestham.
Eng: SSL/TLS secures communication between viewers and CloudFront using HTTPS.

10.What are CloudFront Functions?
CloudFront Functions are lightweight code functions that run at CloudFront edge locations. They are useful for simple request and response transformations with very low latency.
Simple ga gurthupettuko:
CloudFront Functions ante edge location lo small code run chesi simple request or response changes cheyyadaniki use chestham.
Eng: CloudFront Functions run lightweight code at CloudFront edge locations.

11.What is Lambda@Edge?
Lambda@Edge allows Lambda functions to run at CloudFront edge locations in response to CloudFront events. It is useful for more advanced request and response processing.
Simple ga gurthupettuko:
Lambda@Edge ante CloudFront edge locations lo Lambda code run chesi advanced processing cheyyadaniki use chestham.
Eng: Lambda@Edge runs Lambda functions at CloudFront edge locations.

12.What is the difference between CloudFront Functions and Lambda@Edge?
CloudFront Functions are designed for lightweight and fast edge processing, while Lambda@Edge provides more advanced processing capabilities. They have different runtime, event, and resource limitations.
Simple ga gurthupettuko:
CloudFront Functions simple and lightweight tasks kosam, Lambda@Edge advanced processing kosam use chestham.
Eng: CloudFront Functions are lightweight, while Lambda@Edge supports more advanced processing.

13.What is an Edge Location?
An Edge Location is a location in the CloudFront global network where content can be cached and delivered closer to users. It helps reduce latency when users access content.
Simple ga gurthupettuko:
Edge Location ante users ki daggaraga content cache chesi fast ga deliver chese CloudFront location.
Eng: An Edge Location delivers cached content closer to users.

14.What is CloudFront Caching?
CloudFront Caching stores frequently requested content at edge locations. When users request cached content, CloudFront can serve it from the edge location instead of requesting it from the origin every time.
Simple ga gurthupettuko:
Caching ante frequently requested content ni edge lo store chesi fast ga serve cheyyadam.
Eng: CloudFront Caching stores content at edge locations for faster delivery.

15.What is a Cache Hit?
A Cache Hit occurs when CloudFront finds the requested object in its cache and serves it directly to the viewer without contacting the origin.
Simple ga gurthupettuko:
Cache Hit ante requested content CloudFront cache lo already unte direct ga serve cheyyadam.
Eng: A Cache Hit occurs when CloudFront serves content directly from its cache.

16.What is a Cache Miss?
A Cache Miss occurs when the requested object is not available in the CloudFront cache. CloudFront then requests the content from the origin and can cache it for future requests.
Simple ga gurthupettuko:
Cache Miss ante content cache lo lekapothe CloudFront origin nundi content teesukuntundi.
Eng: A Cache Miss occurs when CloudFront must retrieve content from the origin.

17.What is a CloudFront Invalidation?
A CloudFront Invalidation removes specific cached objects from CloudFront edge caches before their normal expiration. It is useful when updated content needs to be served immediately.
Simple ga gurthupettuko:
Invalidation ante old cached content ni remove chesi new content immediate ga serve cheyyadaniki use chestham.
Eng: A CloudFront Invalidation removes cached objects so updated content can be served.

18.What is the difference between CloudFront and S3?
S3 is an object storage service used to store files and data, while CloudFront is a CDN used to deliver content to users with low latency. CloudFront can use an S3 bucket as an origin.
Simple ga gurthupettuko:
S3 data store chestundi, CloudFront stored content ni users ki fast ga deliver chestundi.
Eng: S3 stores data, while CloudFront delivers content with low latency.

19.What is the difference between CloudFront and a Load Balancer?
CloudFront is a CDN that caches and delivers content from edge locations, while a Load Balancer distributes incoming traffic across backend targets.
Simple ga gurthupettuko:
CloudFront content delivery and caching kosam, Load Balancer backend traffic distribution kosam use chestham.
Eng: CloudFront delivers cached content, while a Load Balancer distributes traffic across targets.

20.Can CloudFront use an S3 bucket as an origin?
Yes. CloudFront can use an S3 bucket as an origin to deliver stored objects to users. This is commonly used to securely and efficiently deliver static content.
Simple ga gurthupettuko:
CloudFront ki S3 bucket ni origin ga use chesi files ni users ki fast ga deliver cheyyachu.
Eng: CloudFront can use an S3 bucket as an origin for content delivery.

21.Can CloudFront use an Application Load Balancer as an origin?
Yes. CloudFront can use an Application Load Balancer as an origin. This is useful when CloudFront provides global content delivery and the ALB routes requests to backend applications.
Simple ga gurthupettuko:
CloudFront origin ga ALB ni use chesi global users nundi traffic ni application ki deliver cheyyachu.
Eng: CloudFront can use an Application Load Balancer as an origin.

22.How does CloudFront work?
The user sends a request to CloudFront. CloudFront checks the nearest suitable edge location for cached content and serves it if available. If the content is not cached, CloudFront gets it from the origin, returns it to the user, and can cache it for future requests.
Simple ga gurthupettuko:
User → CloudFront Edge → Cache Hit ayite direct response, Cache Miss ayite Origin nundi content teesukuntundi.
Eng: CloudFront serves cached content from edge locations or retrieves it from the origin when needed.

23.How would you troubleshoot a CloudFront issue?
I would check the distribution status, origin configuration, behaviors, cache policies, origin request policies, DNS configuration, SSL/TLS certificate, permissions, and CloudFront logs. I would also check whether the origin is healthy and accessible.
Simple ga gurthupettuko:
CloudFront problem vaste Distribution → Origin → Behaviors → Cache → DNS → SSL/TLS → Permissions check chestham.
Eng: For a CloudFront issue, I would check the distribution, origin, behaviors, caching, DNS, SSL/TLS, and permissions.

24.Why do we use CloudFront?
We use CloudFront to deliver content globally with low latency, reduce the load on origins through caching, and improve application performance and availability.
Simple ga gurthupettuko:
Global users ki content fast ga deliver cheyyadaniki and origin load reduce cheyyadaniki CloudFront use chestham.
Eng: We use CloudFront for low-latency content delivery and to reduce origin load.

25.What is the most important thing to remember about CloudFront?
CloudFront is an AWS CDN that delivers content from edge locations closer to users. The important concepts are Distributions, Origins, Behaviors, Cache Policies, Origin Request Policies, Response Headers Policies, SSL/TLS, CloudFront Functions, and Lambda@Edge.
Simple ga gurthupettuko:
CloudFront ante content ni edge locations nundi users ki fast ga deliver chese AWS CDN service.
Eng: CloudFront is an AWS CDN used to deliver content quickly from edge locations.